Effect of asteroid gardening on extraterrestrial organic matter: implications for the RNA World Hypothesis

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The RNA World Hypothesis, in which life began with small RNA-like molecules, is one plausible model for the origin of life on Earth [1]. Ribose, nucleobases, and other prebiotically relevant molecules found in meteorite extracts point to extraterrestrial delivery as an important source of organics on the early Earth [2,3]. However, ribonucleosides and ribonucleotides, the monomers of RNA, have not been detected in extraterrestrial samples, and their formation from sugar plus nucleobase has not been observed under prebiotic conditions. Here we evaluate whether nucleobases, sugars, and amino acids found in extraterrestrial samples are degradation products of more complex precursors such as ribonucleosides and peptides. The mechanochemically induced reactions driven by asteroid gardening are poorly understood and could contribute to the diversity of organic matter found in meteorites and asteroid samples.

The canonical ribonucleosides (guanosine, adenosine, cytidine, uridine) were milled in a Retsch® MM 400 (30 Hz, 90 min to 6 h) with salts ($FeSO_4$, $NiCl_2$, $AlCl_3$, Na_2CO_3), pure silica, the lunar analogue JSC-1A and crushed Allende CV3 powder [5,6]. Degradation occurred in every system, the main product being the corresponding nucleobase (guanine, adenine, cytosine, uracil), each already detected in Murchison and Bennu [2,4], whereas pre-heated standards remained intact. Ongoing work addresses other mechanochemical devices and organic-mineral interactions, with emphasis on phyllosilicates, one of the major components of carbonaceous chondrite matrices. This asteroid gardening is also being modeled to link the pressures generated in the mixer mill with those measured in extraterrestrial samples.

These results indicate that mechanical energy can alter the soluble organic matter of carbonaceous asteroids and meteorites, changing the diversity of molecules delivered to the surfaces of habitable worlds.

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